



Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
Institut Teknologi Sepuluh Nopember

Laporan Akhir

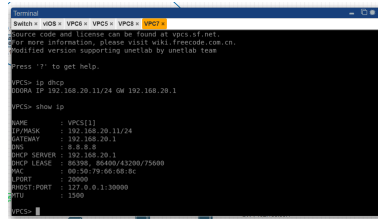
Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

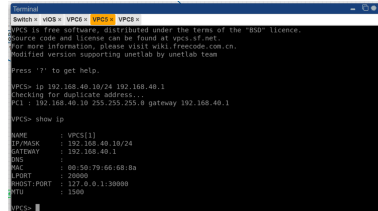
Alya Hasna Fadilah - 5024241044

2026

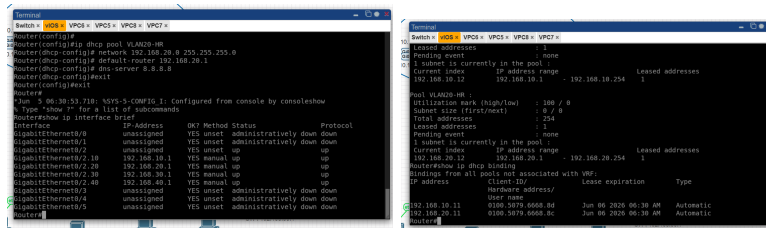
3. Konfigurasi DHCP-Server untuk Vlan 20 ip dhcp excluded-address 192.168.20.1 192.168.20.10



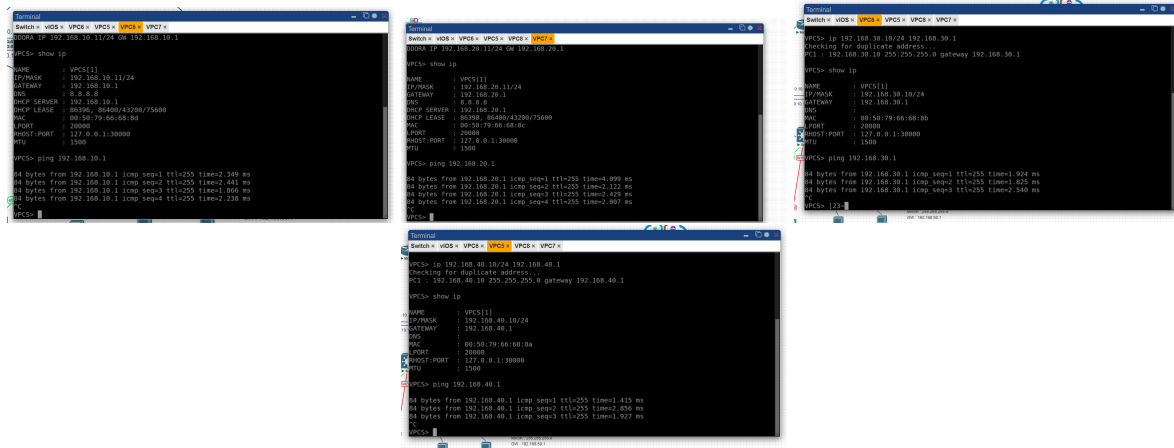
4. Konfigurasi IP Static pada VPCS untuk vlan 30 dan 40



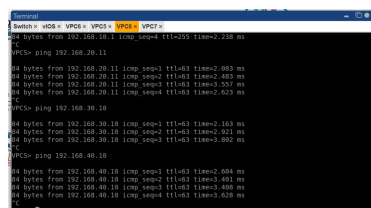
5. Verifikasi Cisco Router



6. Pengujian dari VPCS



7. Uji ping antar-Vlan dari vlan 10:



3. Mengiklankan seluruh network Huawei ke area OSPF

4. Mengatur nilai OSPF cost pada interface Huawei router

5. Konfigurasi OSPF pada MikroTik, Mengiklan kan network yang terhubung ke mikrotik ke area backbone OSPF

6. Mengatur nilai OSPF cost pada interface mikrotik sebagai jalur cadangan.

4

1 Analisis Hasil Percobaan

1.1 Konfigurasi Cisco Switch

Pada praktikum pertama dilakukan konfigurasi VLAN, access port, dan trunk port pada Cisco Switch. Berdasarkan hasil verifikasi, VLAN 10 (Finance), VLAN 20 (HR), VLAN 30 (IT), dan VLAN 40 (Marketing) berhasil dibuat serta berada dalam kondisi aktif. Selain itu, setiap port yang terhubung ke perangkat klien telah ditempatkan pada VLAN yang sesuai sehingga segmentasi jaringan dapat diterapkan dengan baik.

Pemeriksaan menggunakan perintah `show interfaces trunk` menunjukkan bahwa koneksi trunk berhasil meneruskan trafik VLAN 10, 20, 30, dan 40 menuju Cisco Router. Hasil tersebut membuktikan bahwa mekanisme trunking telah berjalan dengan benar.

Keberhasilan konfigurasi dapat dilihat dari kemampuan perangkat dalam VLAN yang sama untuk saling berkomunikasi. Sebaliknya, perangkat yang berada pada VLAN berbeda belum dapat berkomunikasi karena belum tersedia perangkat Layer 3 yang menjalankan fungsi routing. Kondisi ini sesuai dengan konsep dasar VLAN yang memisahkan broadcast domain sehingga komunikasi antar VLAN memerlukan mekanisme inter-VLAN routing.

1.2 Konfigurasi Cisco Router

Pada praktikum kedua Router-on-a-Stick (ROAS) menggunakan Cisco Router yang berfungsi sebagai gateway bagi seluruh VLAN sekaligus sebagai DHCP Server untuk VLAN 10 dan VLAN 20. Hasil verifikasi menunjukkan bahwa seluruh subinterface berhasil dibuat dan telah diberikan alamat IP yang sesuai sebagai gateway masing-masing VLAN.

Perangkat klien pada VLAN 10 dan VLAN 20 berhasil memperoleh alamat IP secara otomatis melalui layanan DHCP. Hal ini menunjukkan bahwa layanan DHCP telah berjalan dengan baik. Hasil menunjukkan bahwa seluruh klien dapat terhubung ke gateway masing-masing. Selain itu, komunikasi antar VLAN juga berhasil dilakukan. Hasil tersebut membuktikan bahwa fungsi inter-VLAN routing telah berjalan dengan baik, di mana router bertindak sebagai perangkat Layer 3 yang meneruskan paket antar jaringan yang berbeda.

1.3 Konfigurasi Huawei Router dan IP Address Antar Router

Pada praktikum ketiga dilakukan konfigurasi alamat IP pada Huawei Router, Cisco Router, dan MikroTik Router untuk membangun konektivitas antar perangkat jaringan. Setiap koneksi point-to-point.

Hasil pengujian menunjukkan bahwa seluruh interface berada dalam kondisi aktif up dan setiap router dapat berkomunikasi antar router tetangganya. Selain itu, perangkat pada LAN Huawei juga berhasil terhubung dengan gateway Huawei Router pada jaringan 192.168.50.0/24.

Beberapa masalah yang bisa terjadi pada tahap ini kesalahan subnet mask, penggunaan alamat IP yang tidak sesuai dengan tabel addressing, atau interface yang belum diaktifkan menggunakan perintah `no shutdown`. Berdasarkan hasil yang diperoleh, seluruh koneksi antar router berhasil dibangun.

1.4 Konfigurasi OSPF Multivendor dan OSPF Cost

Pada praktikum keempat dilakukan implementasi protokol routing dinamis OSPF pada perangkat Cisco, Huawei, dan MikroTik. Tujuan utama kegiatan ini adalah membangun pertukaran informasi routing secara otomatis pada lingkungan jaringan yang menggunakan perangkat dari vendor yang berbeda.

Hasil menunjukkan bahwa seluruh router berhasil membentuk hubungan neighbor OSPF. Cisco Router dapat mengenali Huawei Router dan MikroTik Router sebagai neighbor, begitu pula sebaliknya. Kondisi ini menunjukkan bahwa proses pertukaran Hello Packet dan pembentukan adjacency OSPF berlangsung dengan baik.

Selain itu, tabel routing pada masing-masing router berhasil memperoleh informasi rute secara dinamis dari jaringan lain. Cisco Router memperoleh rute menuju jaringan LAN Huawei, Huawei Router mendapatkan rute menuju VLAN pada Cisco, sedangkan MikroTik Router menerima informasi routing dari kedua router lainnya.

Pengujian konektivitas end-to-end menunjukkan bahwa perangkat pada VLAN Cisco dapat berkomunikasi dengan LAN Huawei, begitu pula sebaliknya. Hasil tersebut membuktikan bahwa OSPF mampu menentukan jalur terbaik menggunakan Shortest Path First (SPF) berdasarkan nilai cost yang diberikan pada setiap interface.

1.5 Pengujian Failover OSPF

Pada praktikum kelima dilakukan pengujian mekanisme failover OSPF untuk mengevaluasi kemampuan jaringan dalam mempertahankan konektivitas ketika terjadi gangguan pada jalur utama. Pengujian dilakukan dengan menonaktifkan link utama secara bertahap dan mengamati perubahan rute yang dipilih oleh OSPF.

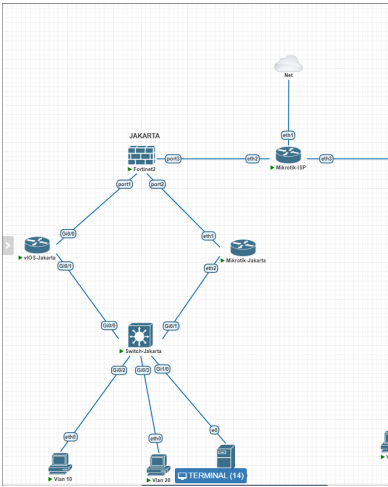
Pada kondisi normal, trafik jaringan melewati jalur utama antara Cisco Router dan Huawei Router sesuai nilai cost yang telah ditentukan. Ketika salah satu link utama dinonaktifkan, OSPF tetap dapat mempertahankan konektivitas melalui jalur utama yang masih tersedia. Setelah kedua link utama terputus, OSPF secara otomatis melakukan proses konvergensi dan mengalihkan trafik melalui MikroTik Router sebagai jalur cadangan.

Hasil pengujian menunjukkan bahwa jaringan tetap dapat beroperasi meskipun terjadi kegagalan pada jalur utama. Hal ini membuktikan bahwa OSPF memiliki kemampuan untuk mendeteksi perubahan topologi jaringan serta memilih rute alternatif secara otomatis tanpa memerlukan konfigurasi ulang secara manual.

Setelah seluruh link utama diaktifkan kembali, OSPF melakukan pembaruan tabel routing dan mengembalikan trafik ke jalur utama yang memiliki nilai cost lebih rendah. Dengan demikian, mekanisme redundansi jaringan dan failover OSPF berhasil dibuktikan sesuai teori yang dipelajari.

2 Hasil Tugas Modul

2.1 TUMOD 1



Gambar 1: Screenshot topologi Jakarta

```
Switch-Jakarta> Switch-Surabaya> vIOS-Jakarta> Mikrotik-Jakarta> Ubuntu-Vlan 80> Mikrotik-ISP> Fortinet2>
Vlan 10> Vlan 20> Mikrotik-Surab...> Fortinet2> Vlan 30> Vlan 40> Ttyouare-Vlan 40>

Press RETURN to get started.

*****
* IOSv is strictly limited to use for evaluation, demonstration and IOS *
* education. IOSv is provided as-is and is not supported by Cisco's *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****
Switchenable
Switchshow vlan br

VLAN Name                Status    Ports
-----
1  default                active    Gi1/1, Gi1/2, Gi1/3
10  FINANCE                 active    Gi0/2
20  IT                      active    Gi0/3
60  SERVER-HQ              active    Gi1/0
1002 fdci-default         act/unsup
1003 token-ring-default   act/unsup
1004 fdginet-default       act/unsup
1005 trnet-default         act/unsup
Switch#
```

Gambar 2: Screenshot show vlan brief

```
Switch-Jakarta> Switch-Surabaya> vIOS-Jakarta> Mikrotik-Jakarta> Ubuntu-Vlan 80> Mikrotik-ISP> Fortinet2>
Vlan 10> Vlan 20> Mikrotik-Surab...> Fortinet2> Vlan 30> Vlan 40> Ttyouare-Vlan 40>

Switchenable
Switchshow vlan br

VLAN Name                Status    Ports
-----
1  default                active    Gi1/1, Gi1/2, Gi1/3
10  FINANCE                 active    Gi0/2
20  IT                      active    Gi0/3
60  SERVER-HQ              active    Gi1/0
1002 fdci-default         act/unsup
1003 token-ring-default   act/unsup
1004 fdginet-default       act/unsup
1005 trnet-default         act/unsup
Switchshow interfaces trunk

Port      Mode      Encapsulation  Status  Native vlan
Gi0/0     on        802.1q         trunking  1
Gi0/1     on        802.1q         trunking  1

Port      Vlans allowed on trunk
Gi0/0     10,20,60
Gi0/1     10,20,60

Port      Vlans allowed and active in management domain
Gi0/0     10,20,60
Gi0/1     10,20,60

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,60
Gi0/1     10,20,60
Switch#
```

Gambar 3: Screenshot show interfaces trunk

2.2 TUMOD 2

```
Switch-Jakarta > Switch-Surabaya > viOS-Jakarta > Mikrotik-Jakarta > Ubuntu-Vlan 10 > Mikrotik-ISP > Fortinet2 >
Vlan 10 > Vlan 20 > Mikrotik-Surabaya > Fortinet2 > Vlan 30 > Vlan 40 > Tinycore-Vlan 40 >

Press RETURN to get started.

*****
* IOSv is strictly limited to use for evaluation, demonstration and IOS *
* education. IOSv is provided as-is and is not supported by Cisco's *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****

Router#enable
Router#show ip interface brief
Interface IP-Address OK? Method Status P
GigabitEthernet0/0 10.10.100.2 YES manual up
GigabitEthernet0/1 unassigned YES unset up
GigabitEthernet0/1.10 192.168.10.2 YES manual up
GigabitEthernet0/1.20 192.168.20.2 YES manual up
GigabitEthernet0/1.60 192.168.60.2 YES manual up
GigabitEthernet0/2 unassigned YES unset administratively down do
GigabitEthernet0/3 unassigned YES unset administratively down do
Router#
```

Gambar 4: Screenshot show ip interface brief.

```
Switch-Jakarta > Switch-Surabaya > viOS-Jakarta > Mikrotik-Jakarta > Ubuntu-Vlan 10 > Mikrotik-ISP > Fortinet2 > Vlan 10 > Vlan 20 >
Mikrotik-Surabaya > Fortinet2 > Vlan 30 > Vlan 40 > Tinycore-Vlan 40 >

*****
* IOSv is strictly limited to use for evaluation, demonstration and IOS *
* education. IOSv is provided as-is and is not supported by Cisco's *
* Technical Advisory Center. Any use or disclosure, in whole or in part, *
* of the IOSv Software or Documentation to any third party for any *
* purposes is expressly prohibited except as otherwise authorized by *
* Cisco in writing. *
*****

Router#enable
Router#show ip interface brief
Interface IP-Address OK? Method Status P
GigabitEthernet0/0 10.10.100.2 YES manual up
GigabitEthernet0/1 unassigned YES unset up
GigabitEthernet0/1.10 192.168.10.2 YES manual up
GigabitEthernet0/1.20 192.168.20.2 YES manual up
GigabitEthernet0/1.60 192.168.60.2 YES manual up
GigabitEthernet0/2 unassigned YES unset administratively down do
GigabitEthernet0/3 unassigned YES unset administratively down do
Router#show vrrp brief
Interface Grp Pri Time Own Pre State Master addr Group addr
Gi0/1.10 10 120 3531 Y Master 192.168.10.2 192.168.10.1
Gi0/1.20 20 90 3648 Y Backup 192.168.20.3 192.168.20.1
Gi0/1.60 60 120 3531 Y Master 192.168.60.2 192.168.60.1
Router#
```

Gambar 5: Screenshot show vrrp brief.

```
Switch-Jakarta > Switch-Surabaya > viOS-Jakarta > Mikrotik-Jakarta > Ubuntu-Vlan 10 > Mikrotik-ISP > Fortinet2 > Vlan 10 > Vlan 20 >
Mikrotik-Surabaya > Fortinet2 > Vlan 30 > Vlan 40 > Tinycore-Vlan 40 >

!
interface GigabitEthernet0/1.10
 encapsulation dot1q 10
 ip address 192.168.10.2 255.255.255.0
 ip helper-address 192.168.60.10
 vrrp 10 ip 192.168.10.1
 vrrp 10 priority 120
end

Router#show running-config interface gi0/1.20
Building configuration...

Current configuration : 181 bytes
!
interface GigabitEthernet0/1.20
 encapsulation dot1q 20
 ip address 192.168.20.2 255.255.255.0
 ip helper-address 192.168.60.10
 vrrp 20 ip 192.168.20.1
 vrrp 20 priority 90
end

Router#show running-config interface gi0/1.60
Building configuration...

Current configuration : 149 bytes
!
interface GigabitEthernet0/1.60
 encapsulation dot1q 60
 ip address 192.168.60.2 255.255.255.0
 vrrp 60 ip 192.168.60.1
 vrrp 60 priority 120
end
```

Gambar 6: Screenshot konfigurasi subinterface.

```
Router#ping 10.10.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Router#
```

Gambar 7: Screenshot ping dari Cisco Router ke FortiGate Jakarta.

2.3 TUMOD 3

```
admin@mikrotik > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 192.168.10.1/24 192.168.10.0 vlan10-finance
1 192.168.20.1/24 192.168.20.0 vlan20-it
2 192.168.60.1/24 192.168.60.0 vlan60-server
3 10.10.101.2/24 10.10.101.0 ether1
4 1 192.168.10.1/32 192.168.10.1 vrrp20
5 192.168.20.1/32 192.168.20.1 vrrp20
6 1 192.168.60.1/32 192.168.60.1 vrrp60
```

Gambar 8: Screenshot /ip address print.

```
admin@mikrotik > /interface vrrp print
Flags: X - disabled, I - invalid, R - running, M - master, B - backup
# NAME INTF... MAC-ADDRESS VRI PRI INTERVAL V V3...
0 B vrrp20 vlan... 00:80:5E:00:01:0A 10 90 1s 2 ipv4
1 B vrrp20 vlan... 00:80:5E:00:01:1A 20 120 1s 2 ipv4
2 B vrrp60 vlan... 00:80:5E:00:01:3C 60 90 1s 2 ipv4
```

Gambar 9: Screenshot /interface vrrp print.

```
admin@mikrotik > /ip dhcp-relay print
Flags: X - disabled, I - invalid
# NAME INTERFACE DHCP-SERVER LOCAL-ADDRESS
0 XI relay1 vlan10-finance 192.168.60.10 192.168.10.3
1 XI relay2 vlan20-it 192.168.60.10 192.168.20.3
```

Gambar 10: Screenshot /ip dhcp-relay print.

```
admin@mikrotik > /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
D - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 A 0.0.0.0/0 10.10.101.1 1
1 ADC 10.10.101.0/30 10.10.101.2 ether1 0
2 ADC 192.168.10.0/24 192.168.10.3 vlan10-finance 0
3 ADC 192.168.20.0/24 192.168.20.3 vlan20-it 0
4 ADC 192.168.20.1/32 192.168.20.1 vrrp20 0
5 ADC 192.168.60.0/24 192.168.60.3 vlan60-server 0
```

Gambar 11: Screenshot /ip route print.

```
admin@mikrotik > ping 10.10.101.1
500 ms
0 10.10.101.1 56 255 ms
1 10.10.101.1 56 255 ms
2 10.10.101.1 56 255 ms
3 10.10.101.1 56 255 ms
4 10.10.101.1 56 255 ms
5 10.10.101.1 56 255 ms
sent=6 received=6 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=3ms
```

Gambar 12: Screenshot ping dari Mikrotik ke FortiGate Jakarta.

2.4 TUMOD 4

```
root@km:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link-loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 50:44:22:01:9f:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.60.10/24 brd 192.168.60.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::5244:22ff:fe01:9f00/64 scope link
        valid_lft forever preferred_lft forever
root@km:~#
```

Gambar 13: Screenshot ip a.

```
root@km:~# ip route
default via 192.168.60.1 dev eth0 proto static
192.168.60.0/24 dev eth0 proto kernel scope link src 192.168.60.10
root@km:~#
```

Gambar 14: Screenshot ip route.

```

root@kvm:~# sudo cat /etc/dhcp/dhcpd.conf
authoritative;

default-lease-time 600;
max-lease-time 7200;

option domain-name-servers 8.8.8.8, 1.1.1.1;

subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.100 192.168.10.200;
    option routers 192.168.10.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.10.255;
}

subnet 192.168.20.0 netmask 255.255.255.0 {
    range 192.168.20.100 192.168.20.200;
    option routers 192.168.20.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.20.255;
}

subnet 192.168.60.0 netmask 255.255.255.0 {
    option routers 192.168.60.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.60.255;
}
root@kvm:~# _

```

Gambar 15: Screenshot isi file /etc/dhcp/dhcpd.conf.

```

root@kvm:~# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=110 time=25.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=110 time=25.6 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=110 time=26.0 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 25.579/25.812/26.047/0.191 ms
root@kvm:~# _

```

Gambar 16: Screenshot ping 8.8.8.8.

2.5 TUMOD 5

```
Switch-Jakarta x Switch-Sura... x vIOS-Jakarta x Mikrotik-Jak... x Ubuntu-Vlan... x Mikrotik-ISP x Fortinet2 x Vlan 10 x Vlan 20 x
Mikrotik-Sur... x Fortinet3 x Vlan 30 x Vlan 40 x Tinycore-Via... x
FortiGate-VM64-KVM # Timeout

FortiGate-VM64-KVM login: admin
Password:
Welcome!

FortiGate-VM64-KVM # get system interface physical
== [onboard]
==[port1]
mode: static
ip: 10.10.100.1 255.255.255.252
ipv6: ::/0
status: up
speed: 10000Mbps (Duplex: full)
FEC: none
FEC_cap: none
==[port2]
mode: static
ip: 10.10.101.1 255.255.255.252
ipv6: ::/0
status: up
speed: 10000Mbps (Duplex: full)
FEC: none
FEC_cap: none
==[port3]
mode: static
ip: 10.0.12.2 255.255.255.252
ipv6: ::/0
status: up
speed: 10000Mbps (Duplex: full)
FEC: none
--More--
```

Gambar 17: Screenshot get system interface physical.

```
FortiGate-VM64-KVM # get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter are
V - BGP VPNv4
* - candidate default

Routing table for VRF=0
* 0.0.0.0/0 [10/0] via 10.0.12.1, port3, [1/0]
10.0.12.0/30 is directly connected, port3
10.10.100.0/30 is directly connected, port1
10.10.101.0/30 is directly connected, port2
172.16.0.1/32 is directly connected, GRE-JKT-SBY
172.16.0.2/32 is directly connected, GRE-JKT-SBY
192.168.10.0/24 [10/0] via 10.10.100.2, port1, [1/0]
[10/0] via 10.10.101.2, port2, [1/0]
192.168.20.0/24 [10/0] via 10.10.100.2, port1, [1/0]
[10/0] via 10.10.101.2, port2, [1/0]
E2 192.168.30.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:31:10, [1/0]
E2 192.168.40.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:31:10, [1/0]
192.168.60.0/24 [10/0] via 10.10.100.2, port1, [1/0]
[10/0] via 10.10.101.2, port2, [1/0]
```

Gambar 18: Screenshot get router info routing-table all

```
Switch-Jakarta x Switch-Sura... x vIOS-Jakarta x Mikrotik-Jak... x Ubuntu-Vlan... x Mikrotik-ISP x Fortinet2 x Vlan 10 x Vlan 20 x Mikr
edit 1
set name "VLAN-to-INTERNET"
set uuid 796c0b12-6504-51f1-b7bf-5abaead7acce
set srcintf "port2" "port1"
set dstintf "port3"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
set nat enable
next
edit 2
set name "LAN-to-GRE"
set uuid 010a523c-6509-51f1-84ce-e4c190359657
set srcintf "port1" "port2"
set dstintf "GRE-JKT-SBY"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
next
edit 3
set name "GRE-to-LAN"
set uuid 1892c858-6509-51f1-f5a6-51dd8fa54dba
set srcintf "GRE-JKT-SBY"
set dstintf "port1" "port2"
set action accept
set srcaddr "all"
set dstaddr "all"
set schedule "always"
set service "ALL"
next
```

Gambar 19: Screenshot firewall policy.

```

fortiGate-VM64-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=112 time=22.9 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=22.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=22.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=23.0 ms
^C
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 22.3/22.6/23.0 ms

```

Gambar 20: Screenshot ping 8.8.8.8

```

FortiGate-VM64-KV1# execute ping 172.16.0.2
PING 172.16.0.2 (172.16.0.2): 56 data bytes
56 bytes from 172.16.0.2: icmp_seq=0 ttl=255 time=2.9 ms
56 bytes from 172.16.0.2: icmp_seq=1 ttl=255 time=1.6 ms
56 bytes from 172.16.0.2: icmp_seq=2 ttl=255 time=1.2 ms
^C
--- 172.16.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.2/1.9/2.9 ms

```

Gambar 21: Screenshot ping ke IP tunnel Surabaya

```
FortiGate-VM64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID      Pri   State           Dead Time   Address        Interface
2.2.2.2          1    Full/ -         00:00:37   172.16.0.2    GRE-JKT-SBY
```

Gambar 22: Screenshot get router info ospf neighbor

```
FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
0 E2    192.168.30.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]
0 E2    192.168.40.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]
```

Gambar 23: Screenshot get router info routing-table ospf

2.6 TUMOD 6

```
Switch-Jakarta > Switch-Sura... > vIOS-Jakarta > Mikrotik-Jak... > Ubuntu-Vlan... > Mikrotik-ISP > Forti...
Mikrotik-Sur... > Fortinet > Vlan 30 > Vlan 40 > Tinycore-Via... >

[admin@Mikrotik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 10.4.89.179/24 10.4.89.0 ether1
1 10.0.12.1/30 10.0.12.0 ether2
2 10.0.13.1/30 10.0.13.0 ether3
[admin@Mikrotik] >
```

Gambar 24: Screenshot /ip address print hasil di ether 1 bisa saja berbeda karena emang dinamic.

```
[admin@Mikrotik] > /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADS 0.0.0.0/0 10.4.89.1 1
1 ADC 10.0.12.0/30 10.0.12.1 ether2 0
2 ADC 10.0.13.0/30 10.0.13.1 ether3 0
3 ADC 10.4.89.0/24 10.4.89.170 ether1 0
```

Gambar 25: Screenshot /ip route print.

```
[admin@Mikrotik] > /ip firewall nat print
Flags: X - disabled, I - invalid, D - dynamic
0 chain=srcnat action=masquerade out-interface=ether1
```

Gambar 26: Screenshot /ip firewall nat print.

```
[admin@Mikrotik] > ping 8.8.8.8
SEQ HOST SIZE TTL TIME STATUS
0 8.8.8.8 56 113 2ms
1 8.8.8.8 56 113 2ms
2 8.8.8.8 56 113 2ms
icmp=3 received=3 packet-loss=0% min-rtt=2ms avg-rtt=2ms max-rtt=2ms
```

Gambar 27: Screenshot ping 8.8.8.8.

```
FortiGate-VM64-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.3 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=1.0 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=1.1 ms
^C
-- 10.0.12.2 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.0/1.1/1.3 ms

FortiGate-VM64-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.5 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=1.1 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=0.6 ms
^C
-- 10.0.12.2 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/3.7/8.6 ms
```

Gambar 28: Screenshot ping antar-WAN FortiGate.

2.7 TUMOD 7

```
Switch-Jakarta > Switch-Surabaya > VIDS-Jakarta > MikroTik-Jak... > Ubuntu-Vlan... > Mikrotik-ISP > Fortinet2 > Vlan10 > Vlan20 >
Mikrotik-Sur... > Fortinet2 > Vlan20 > Vlan40 > Tinycore-Vlan... >

Press RETURN to get started.

*****
IOSv is strictly limited to use for evaluation, demonstration and IOS
education. IOSv is provided as-is and is not supported by Cisco's
Technical Advisory Center. Any use or disclosure, in whole or in part,
of the IOSv Software or Documentation to any third party for any
purposes is expressly prohibited except as otherwise authorized by
Cisco in writing.
*****
Switchenable
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1  default                active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
10  SALES                  active    Gi0/1
100  OPERATIONS             active    Gi0/2, Gi0/3
1002 fddi-default           act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
Switch#
```

Gambar 29: Screenshot show vlan brief.

```
Switch#show interfaces trunk

Port      Mode          Encapsulation  Status        Native vlan
-----
Gi0/0     on            802.1q         trunking      1
Gi0/0     30,40         Vlan allowed on trunk
Gi0/0     30,40         Vlan allowed and active in management domain
Gi0/0     30,40         Vlan in spanning tree forwarding state and not pruned
Switch#
```

Gambar 30: Screenshot show interfaces trunk.

```
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 192.168.30.1/24 192.168.30.0 vlan30-sales
1 192.168.40.1/24 192.168.40.0 vlan40-operations
2 10.10.200.2/30 10.10.200.0 ether1
```

Gambar 31: Screenshot /ip address print.

```
[admin@MikroTik] > /ip dhcp-server print
Flags: D - dynamic, X - disabled, I - invalid
# NAME INTERFACE RELAY ADDRESS-POOL LEASE-TIME ADD-ARP
0 dhcp30 vlan30-sales pool30 10m
```

Gambar 32: Screenshot /ip dhcp-server print.

```
[admin@MikroTik] > /ip pool print
# NAME RANGES
0 pool30 192.168.30.100-192.168.30.200
```

Gambar 33: Screenshot /ip pool print.

```
[admin@MikroTik] > /ip route print
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
P - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 A S 8.8.8.8/8 10.10.200.2 10.10.200.1 1
1 ADC 10.10.200.0/30 10.10.200.1 ether1 0
2 ADC 192.168.30.0/24 192.168.30.1 vlan30-sales 0
3 ADC 192.168.40.0/24 192.168.40.1 vlan40-operations 0
```

Gambar 34: Screenshot /ip route print.

```
Switch-Jakarta > Switch-Surabaya > VIDS-Jakarta > MikroTik-Jak... > Ubuntu-Vlan... > Mikrotik-ISP > Fortinet2 > Vlan10 > Vlan20 >
Mikrotik-Sur... > Fortinet2 > Vlan20 > Vlan40 > Tinycore-Vlan... >

MTU : 1500

VPCS> ping 192.168.30.1
192.168.30.1 icmp_seq=1 timeout
^C
VPCS>
VPCS> ip dhcp
DORA IP 192.168.30.200/24 GW 192.168.30.1
VPCS> ping 8.8.8.8
64 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=25.800 ms
64 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=29.143 ms
64 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=25.861 ms
^C
VPCS> ip dhcp
DORA IP 192.168.30.200/24 GW 192.168.30.1
VPCS> show ip
NAME : VPCS[1]
IP/MASK : 192.168.30.200/24
GATEWAY : 192.168.30.1
DNS : 8.8.8.8
DHCP SERVER : 192.168.30.1
DHCP LEASE : 598, 600/300/525
MAC : 08:50:79:66:68:9d
I/PORT : 20000
HOST:PORT : 127.0.0.1:30000
MTU : 1500
VPCS>
```

Gambar 35: Screenshot client VLAN 30 mendapat IP DHCP.

```
VPCS> ping 8.8.8.8
64 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=25.800 ms
64 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=29.143 ms
64 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=25.861 ms
^C
```

Gambar 36: Screenshot ping client Surabaya ke 8.8.8.8.

2.8 TUMOD 8

```

bash-jan@v: ~$ ssh -o StrictHostKeyChecking=no -o UserKnownHostsFile=/dev/null -o LogLevel=quiet -o PasswordAuthentication=no -o Port=2222 root@10.0.13.2
Mar06/20 ~$ PortGate2 Vlan 32 ~ Vlan 42 ~ Tinycore-Via-~
PortGate-VM64-KVM ~ Timeout

PortGate-VM64-KVM login: admin
Password:
Welcome!

PortGate-VM64-KVM # get system interface physical
== [onboard]

mode: static
ip: 10.0.13.2 255.255.255.252
ipw6: :/0
status: up
speed: 1000Mbps (Duplex: full)
FEC: none
FEC_cap: none

== [port2]

mode: static
ip: 10.0.280.1 255.255.255.252
ipw6: :/0
status: up
speed: 1000Mbps (Duplex: full)
FEC: none
FEC_cap: none

== [port3]

mode: static
ip: 0.0.0.0 0.0.0.0
ipw6: :/0
status: down
speed: n/a
FEC: none

~More~

```

Gambar 37: Screenshot get system interface physical

[illegible]

Gambar 38: screenshot get router info routing-table all.

```

config firewall policy
edit 1
    set name "LAN-TO-INTERNET"
    set uuid 1c3ce4bc-6593-51f1-2b6b-03775fdd8f42
    set srcintf "port2"
    set dstintf "port1"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set nat enable
next
edit 2
    set name "LAN-to-GRE"
    set uuid 447f60eb-6599-51f1-25b6-83c7a93f26af
    set srcintf "port2"
    set dstintf "GRE-SBY-JKT"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
next
edit 3
    set name "GRE-to-LAN"
    set uuid 448c7742-6599-51f1-a24f-2ca1bb8f7fb2
    set srcintf "GRE-SBY-JKT"
    set dstintf "port2"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"

```

Gambar 39: Screenshot firewall policy

```
Fortigate-VM64-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=112 time=22.9 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=22.2 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=23.3 ms
^C
-- 8.8.8.8 ping statistics --
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 22.2/22.8/23.3 ms
```

Gambar 40: Screenshot ping ke 8.8.8.8

```
FortiGate-VM64-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=1.2 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=1.1 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=1.2 ms
^C
--- 172.16.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/1.1/1.2 ms
```

Gambar 41: Screenshot ping ke IP tunnel Jakarta.

```
FortiGate-VM64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID    Pri  State           Dead Time   Address      Interface
1.1.1.1        1    Full/ -         00:00:31    172.16.0.1   GRE-SBY-JKT
```

Gambar 42: Screenshot get router info ospf neighbor.

```
FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
O E2   192.168.10.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
O E2   192.168.20.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
O E2   192.168.60.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
```

Gambar 43: Screenshot get router info routing-table ospf.

2.9 TUMOD 9

```
FortiGate-VM64-KVM # execute ping 10.0.12.2
PING 10.0.12.2 (10.0.12.2): 56 data bytes
64 bytes from 10.0.12.2: icmp_seq=0 ttl=254 time=1.5 ms
64 bytes from 10.0.12.2: icmp_seq=1 ttl=254 time=1.1 ms
64 bytes from 10.0.12.2: icmp_seq=2 ttl=254 time=8.6 ms
^C
--- 10.0.12.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.1/3.7/8.6 ms
```

```
FortiGate-VM64-KVM # execute ping 10.0.13.2
PING 10.0.13.2 (10.0.13.2): 56 data bytes
64 bytes from 10.0.13.2: icmp_seq=0 ttl=254 time=1.3 ms
64 bytes from 10.0.13.2: icmp_seq=1 ttl=254 time=1.0 ms
64 bytes from 10.0.13.2: icmp_seq=2 ttl=254 time=1.1 ms
^C
--- 10.0.13.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.0/1.1/1.3 ms
```

Gambar 44: Screenshot ping WAN antar-FortiGate.

```
FortiGate-VM64-KVM # execute ping 172.16.0.2
PING 172.16.0.2 (172.16.0.2): 56 data bytes
64 bytes from 172.16.0.2: icmp_seq=0 ttl=255 time=3.9 ms
64 bytes from 172.16.0.2: icmp_seq=1 ttl=255 time=1.5 ms
64 bytes from 172.16.0.2: icmp_seq=2 ttl=255 time=1.2 ms
^C
--- 172.16.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.2/2.2/3.9 ms
```

```
FortiGate-VM64-KVM # execute ping 172.16.0.1
PING 172.16.0.1 (172.16.0.1): 56 data bytes
64 bytes from 172.16.0.1: icmp_seq=0 ttl=255 time=4.4 ms
64 bytes from 172.16.0.1: icmp_seq=1 ttl=255 time=1.5 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=255 time=1.4 ms
^C
--- 172.16.0.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 1.4/2.4/4.4 ms
```

Gambar 45: Screenshot ping tunnel antar-FortiGate.

```
FortiGate-VM64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID    Pri  State           Dead Time   Address      Interface
2.2.2.2        1    Full/ -         00:00:37    172.16.0.2   GRE-JKT-SBY
```

Gambar 46: Screenshot get router info ospf neighbor.

```

FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
O E2   192.168.30.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]
O E2   192.168.40.0/24 [110/10] via 172.16.0.2, GRE-JKT-SBY, 01:34:16, [1/0]

FortiGate-VM64-KVM # get router info routing-table ospf
Routing table for VRF=0
O E2   192.168.10.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
O E2   192.168.20.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]
O E2   192.168.60.0/24 [110/10] via 172.16.0.1, GRE-SBY-JKT, 01:47:12, [1/0]

```

Gambar 47: Screenshot get router info routing-table ospf.

```

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=60 time=13.084 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=60 time=11.700 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=60 time=10.312 ms
^C
VPCS>

```

Gambar 48: Screenshot ping client Jakarta ke client Surabaya.

```

VPCS> ping 192.168.10.101

84 bytes from 192.168.10.101 icmp_seq=1 ttl=60 time=12.882 ms
84 bytes from 192.168.10.101 icmp_seq=2 ttl=60 time=7.930 ms
84 bytes from 192.168.10.101 icmp_seq=3 ttl=60 time=7.014 ms
^C

```

Gambar 49: Screenshot ping client Surabaya ke client Jakarta.

2.10 TUMOD 10

```

VPCS> ip dhcp
DDORA IP 192.168.10.102/24 GW 192.168.10.1

```

Gambar 50: Screenshot IP DHCP client Jakarta (Vlan 10).

```

VPCS> ip dhcp
DDORA IP 192.168.30.200/24 GW 192.168.30.1

```

Gambar 51: Screenshot IP DHCP client Surabaya (Vlan 20).

```

VPCS> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=28.083 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=25.015 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=28.061 ms
^C
VPCS>

```

Gambar 52: Screenshot ping internet dari Jakarta.

```

VPCS> ping 8.8.8.8

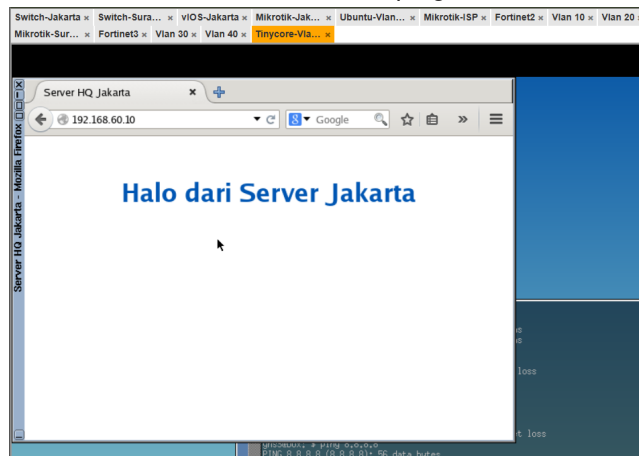
84 bytes from 8.8.8.8 icmp_seq=1 ttl=110 time=25.738 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=110 time=25.469 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=110 time=26.359 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=110 time=26.912 ms
^C

```

Gambar 53: Screenshot ping internet dari Surabaya.

```
VPCS> ping 192.168.40.10
84 bytes from 192.168.40.10 icmp_seq=1 ttl=63 time=8.299 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=63 time=4.425 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=63 time=5.837 ms
VPCS>
```

Gambar 54: Screenshot ping antar-site



Gambar 55: Screenshot akses web server Jakarta dari Surabaya.

```
FortiGate-VM64-KVM # get router info ospf neighbor
OSPF process 0, VRF 0:
Neighbor ID    Pri  State           Dead Time   Address      Interface
1.1.1.1        1    Full/-         00:00:31   172.16.0.1   GRE-SBY-JKT
```

Gambar 56: Screenshot routing table OSPF

```
VPCS> trace 192.168.40.10
Trace to 192.168.40.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.2    11.510 ms  2.193 ms  4.470 ms
 2  10.10.100.1     8.313 ms  3.927 ms  2.594 ms
 3  172.16.0.2      5.450 ms  5.588 ms 11.021 ms
 4  10.10.200.2     5.805 ms  5.475 ms  4.767 ms
 5  *192.168.40.10 11.328 ms (ICMP type:3, code:3, Destination port unreachable)
```

Gambar 57: Analisis singkat jalur traffic Jakarta ke Surabaya.

3 Kesimpulan

Berdasarkan seluruh rangkaian praktikum yang telah dilaksanakan, tujuan pembelajaran pada setiap modul berhasil dicapai dengan baik. Konfigurasi VLAN, access port, dan trunk port dapat diterapkan sehingga segmentasi jaringan berjalan sesuai kebutuhan, sementara implementasi Router-on-a-Stick (ROAS) memungkinkan komunikasi antar VLAN serta distribusi alamat IP secara otomatis melalui DHCP. Selain itu, pengalamatan IP pada perangkat multivendor seperti Cisco, Huawei, dan MikroTik berhasil dilakukan sehingga konektivitas antar perangkat dapat terjalin dengan baik melalui jaringan point-to-point. Penerapan protokol routing dinamis OSPF juga berhasil mendukung pertukaran informasi routing dan komunikasi end-to-end antar jaringan, termasuk pengujian mekanisme failover yang menunjukkan kemampuan jaringan untuk beralih ke jalur cadangan secara otomatis saat terjadi gangguan dan kembali ke jalur utama setelah kondisi normal. Hasil praktikum ini membuktikan kesesuaian antara teori dan implementasi terkait VLAN, inter-VLAN routing, pengalamatan IP, routing dinamis, serta redundansi jaringan.

4 Lampiran



Gambar 58: Dokumentasi Praktikum